The Posterior Coherence Engine: Separating Uncertainty from Incoherence

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October 2026

Abstract

When several sources report on the same question, averaging their reports answers one question: how uncertain is the answer? It cannot answer a second: do the reports admit one coherent model at all? The Posterior Coherence Engine keeps these apart. For a set of local reports it returns (i) the three-level split of predictive uncertainty into disagreement between groups, disagreement within groups and irreducible noise; (ii) a coherence status, either coherent and unique, coherent but not unique (with the width of the set of global laws), or incoherent (with a finite certificate and a consistency radius); and (iii) the Beck–Chevalley defect of a pooling rule, which says when pooling is exact. A decision layer answers, abstains, or flags. On synthetic worlds with known truth, the averaged answers carry no information about incoherence, while the consistency radius detects it with AUROC rising from 0.58 to 0.92 as data per context grow. In a pre-registered test on 2358 items answered by four runs of two model families, keeping where disagreement sits improves the prediction of whether the pooled answer is correct by 0.0175 bits per item (95% interval $[0.0107, 0.0241]$), with the gain concentrated in the two tasks with strong version drift. The incoherence detector cannot be tested on these data, because each item's answers are observed jointly; we say what data would test it.

1 What the engine reports

The inputs are local reports: probability tables on *windows* (sets of variables observed together), and, where available, posteriors from sources arranged in groups. The engine returns three things.

1. Uncertainty about the answer. For groups g with weights w_g and member posteriors q, the chain rule along (group, member) gives exactly

$$H(\rho) \;=\; \underbrace{I(X;\text{group})}_{\text{between}} \;+\; \underbrace{I(X;\text{member} \mid \text{group})}_{\text{within}} \;+\; \underbrace{\mathbb{E}\, H(q)}_{\text{noise}},$$

where ρ is the overall average. Averaging keeps only $H(\rho)$. This is the martingale structure of the posterior tower [1, Thm. 5.3].

2. Coherence of the evidence. Can the window tables be glued into one global law? The engine solves three linear programs.

- The *consistency radius*: the smallest average total-variation adjustment of the window tables that admits a global law.
- If the radius is 0: the range of every atom's probability over all global laws. Width 0 means the global law is unique.

- If the radius is positive: a *certificate*, found by Farkas duality, a functional on window cells that is ≥ 0 on every deterministic assignment (hence on every global law) and negative on the data. For three binary claims seen in pairs it is a cycle (Bell-type) inequality such as $\Phi \leq 2$ [1, Prop. 16.1].

Two overlapping windows always glue; failure needs a cycle in the cover [1, Prop. 16.3].

3. Validity of pooling. For two windows meeting in M, pooling through M (the Markov glue) is exact iff the windows satisfy Beck–Chevalley, with defect c the largest conditional correlation [1, Thm. 14.4]. For a more and a less informed source, the pooling rules at least as good as the better one form a KL ball of radius $I(X; q_j \mid q_i)$ around the Bayes pool, and every rule in it errs at most $\sqrt{2I}$ more often (weighted by margin) than the Bayes decision [2, Thm. 7, Prop. 9].

Decisions. `decide` flags incoherent evidence with its certificate instead of averaging it, flags non-unique reconstructions whose width exceeds a tolerance, and otherwise answers or abstains by a confidence threshold.

2 Using it

```
from pce import Window, coherence, split, decide
import numpy as np
t = np.array([[0.125, 0.375], [0.375, 0.125]]) # each pair agrees w.p. 1/4
dom = {"X0": 2, "X1": 2, "X2": 2}
pairs = [("X0", "X1"), ("X1", "X2"), ("X0", "X2")]
r = coherence(dom, [Window(p, t) for p in pairs])
r.status, r.radius, r.violation # 'incoherent', 0.083, 0.375
s = split([[(1, [0.9, 0.1]), (1, [0.8, 0.2])],
          [(1, [0.2, 0.8]), (1, [0.3, 0.7])]])
s.between, s.within, s.noise # 0.282, 0.012, 0.699 bits
decide(s, r).action # 'flag-incoherent'
```

The same evidence, written as a JSON file, runs from the command line:

```
python -m pce.cli examples/three_token.json
```

Checks. `check_pce.py` (six checks, all passing) verifies that the engine reproduces the paper's three-token model (incoherent iff $\varepsilon < \frac{1}{3}$, fibre width $2\min(a, b)$ above), that every certificate is non-negative on all deterministic assignments and negative on the data (75 incoherent cases among 300 random compatible families), that the split is exact, that the defect matches $|2\varepsilon - 1|/(2\varepsilon)$, that the KL-ball radius matches the companion note, and that the decision layer acts as specified.

3 Synthetic demonstrations

These use worlds with known truth. They debug and illustrate; nothing in the real-data analysis was chosen from them.

Averaging hides conflicts. Random compatible pair tables on a triangle of binary claims, about a quarter of which admit no global law, are observed with n samples per pair. The confidence of the averaged per-claim answers does not distinguish the two classes (AUROC between 0.40 and

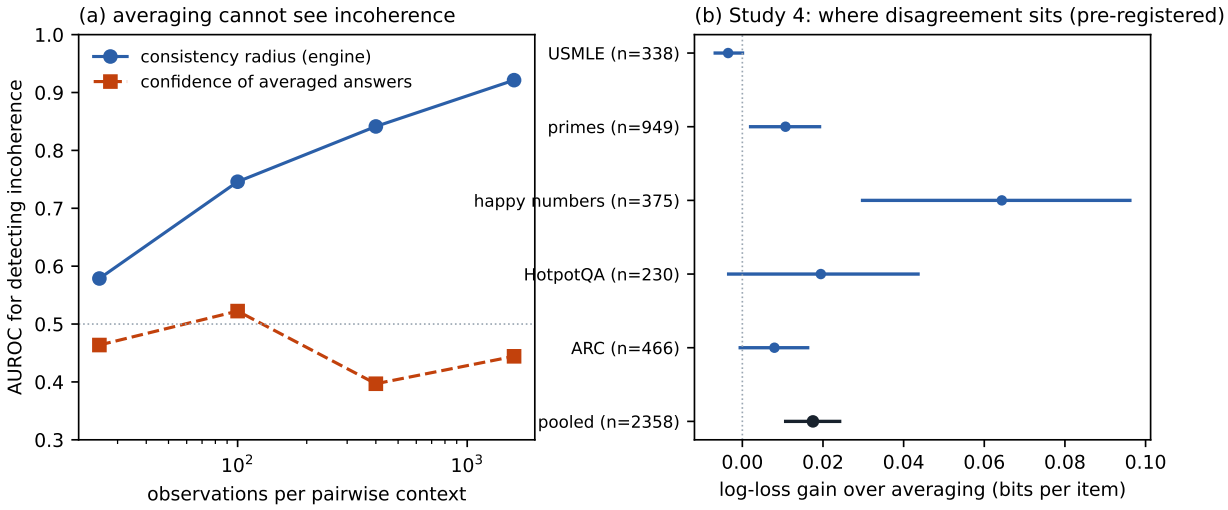


Figure 1: (a) Detecting incoherent evidence on synthetic triangles: the engine's consistency radius against the confidence of averaged answers. (b) Study 4: gain in out-of-fold log loss from adding the between-family term, per task and pooled, with 95% bootstrap intervals. Pre-registered.

0.52); the consistency radius does, with AUROC 0.58, 0.75, 0.84 and 0.92 at $n = 25$, 100, 400 and 1600 (fig. 1a). Many random incoherent families are only barely incoherent, which is why small samples struggle.

Non-unique reconstruction. When only the chain (X_0, X_1), (X_1, X_2) is observed, the engine reports the interval of $P(X_0 = X_2)$ over all global laws. Over 200 random worlds the interval always contains the truth; the single Markov-glue reconstruction never equals it, with median error 0.09 and maximum 0.36.

Where disagreement sits. Items are answered by four runs in two families. In a world with family-specific blind spots, adding the between-family term to (v, H) improves out-of-fold log loss by 0.036 bits ($[0.017, 0.052]$) and AUROC from 0.78 to 0.81. In a null world with exchangeable runs it changes nothing (-0.0004 bits, $[-0.002, 0.001]$). So the test can fail, and does when there is nothing to find.

4 A pre-registered real-data test

Design. The plan was frozen before any outcome was computed (`study4/PREREG_engine.md`, SHA-256 beginning `5c6543bf`). Data: the LLM-drift answers of GPT-4 (March, June) and GPT-3.5 (March, June) on five tasks [3], 2358 items with all four answers and a reference. The pooled answer is the plurality (ties to GPT-4 June). The question is whether the pooled answer is correct. The averaging baseline uses only the pooled vote distribution: vote share v and entropy H. The engine adds the between-family term B. Both are logistic models fitted within task, scored by five-fold out-of-fold log loss.

Result. The pre-registered hypothesis is supported: keeping where disagreement sits improves the decision by 0.0175 bits per item, with interval $[0.0107, 0.0241]$ (table 1). The gain is significant

Table 1: Study 4 (pre-registered). Δ = log-loss gain of the engine over averaging, bits per item; one-sided bootstrap p, Holm-adjusted; AUROC of out-of-fold probabilities.

task	n	accuracy	Δ [95% interval]	p_{Holm}	AUROC: averaging $\rightarrow$ engine
USMLE	338	0.861	-0.0035 [-0.0067, 0.0000]	0.974	0.643 $\rightarrow$ 0.625
primes	949	0.542	0.0107 [0.0020, 0.0191]	0.016	0.873 $\rightarrow$ 0.874
happy numbers	375	0.635	0.0644 [0.0298, 0.0961]	0.005	0.690 $\rightarrow$ 0.842
HotpotQA	230	0.600	0.0195 [-0.0034, 0.0436]	0.100	0.689 $\rightarrow$ 0.707
ARC	466	0.273	0.0080 [-0.0005, 0.0162]	0.091	0.788 $\rightarrow$ 0.790
pooled	2358		**0.0175 [0.0107, 0.0241]**		0.870 $\rightarrow$ 0.875

after Holm correction in two tasks, primes and happy numbers, positive but not significant in HotpotQA and ARC, and slightly negative in USMLE.

Exploratory reading. In primes, 475 of 949 items split two against two *within* each family (the March and June versions of each model disagree), and the pooled answer is right on only 19% of them; in happy numbers the same pattern has accuracy 4% (48 items). Splits *along* family lines are rare and much more often right. At equal total entropy, disagreement within a family over time predicts error more strongly than disagreement between families; the fitted coefficient on B is positive in primes, happy numbers and ARC. The gain is therefore concentrated where model versions drifted, which Chen et al. documented for exactly these tasks. It is a real effect of the information the engine keeps, but it should not be read as a general property of all tasks.

5 What is not yet tested

- *Incoherence on real data.* Each item's answers in Study 4 are observed jointly, and jointly observed tables always admit a global law, so the engine's coherence status is uninformative there. A real test needs evidence gathered in separate contexts, for example asking a model about pairs of related claims in separate prompts. The pairwise-elicitation kit of the main paper (`halluc_study/kit_pairwise_cycles.py`) collects such data; the engine scores it directly.
- *Decisions beyond correctness prediction.* Study 4 measures how well the engine's features predict whether the pooled answer is right. Whether flagging and abstaining by the engine improves a downstream task is the next experiment, and should be pre-registered in the same way.
- *Statistical calibration of the radius.* The radius is a point estimate; a bootstrap or permutation calibration for finite samples, as done for the Beck–Chevalley test in the main paper, would turn it into a test with known error rates.

Code

The package `pce`, the checks, the demonstrations and Study 4 are in the `engine` folder of https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/papers.

References

[1] V. K. Awasthi, Shadows, fibres and secants of the posterior tower. Technical report, October 2026.

[2] V. K. Awasthi, Pooling informed agents: Blackwell order, reconciliation and the KL ball. Companion note, October 2026.

[3] L. Chen, M. Zaharia and J. Zou, How is ChatGPT's behavior changing over time? *Harvard Data Science Review* **6** (2024), no. 2; arXiv:2307.09009. Data: https://github.com/lchen001/LLMDrift.